

Halfway Splits and Finishing Times in the 2023 Boston Marathon: A Simple Linear Regression Analysis of Pacing

Pujitha Bobbili

September 25, 2026

Abstract

Runners are routinely told to start a marathon conservatively, but the size of the penalty for a fast first half is rarely stated in terms a runner can use mid-race. Using official results for 26,526 finishers of the 2023 Boston Marathon, distributed through the SCORE Network Sports Data Repository (Fay, Dykstra, and Ramler 2023), I fit a simple linear regression of net finishing time on half-marathon split time. Each additional minute at halfway is associated with 2.22 additional minutes at the finish (95% confidence interval 2.21 to 2.22), which is reliably more than the two minutes that even pacing would imply, and the halfway split predicts finishing time with a residual standard deviation of about 9.4 minutes. Because the gap widens as the halfway split lengthens, slower runners lose proportionally more time over the second half than faster runners do, and the familiar “double your halfway time” rule understates the finish for nearly nine runners in ten. A more accurate rule for this field is to double the halfway split and add roughly a further fifth of it.

1 Introduction

The half-marathon timing mat is the most visible checkpoint in a marathon. It is where runners take stock of their race, where spectators guess when their friend will appear, and where broadcast commentators project a winning time. The projection almost always used is simple doubling: a runner passing halfway in 1 hour 45 minutes is said to be “on pace” for 3 hours 30 minutes. That rule is exactly right only if a runner covers the second half in the same time as the first, and decades of race results suggest that most runners do not (March et al. 2011; Santos-Lozano et al. 2014). What is less often quantified, in a form that can be applied at the mat, is how much the rule is wrong by, and whether the error is the same for a 2-hour 30-minute runner and a 5-hour runner.

Slowing over the second half of a marathon, known as positive splitting, has a well-documented physiological basis: muscle glycogen depletion, rising core temperature, and accumulated muscle damage all make the closing miles more costly than the opening ones. The extent of the slowdown varies systematically across runners. Slower and less experienced runners fade more than faster ones (March et al. 2011), men fade more than women on average (Deaner et al. 2015), warm conditions amplify the effect, particularly for slower finishers (Ely et al. 2008), and a sizeable minority of recreational runners experience a pronounced late-race collapse (Smyth 2021). Much of this work summarizes pacing by comparing group averages, dividing fields into ability bands or sex groups and reporting mean second-half slowdown within each band.

Comparing group means answers a descriptive question but not a predictive one. A runner at the

mat does not want to know the average slowdown of their ability band; they want a single equation that converts their own split into an expected finishing time, together with an honest statement of how far off that estimate is likely to be. Simple linear regression supplies exactly that, and it has an additional advantage here: the even-pacing rule corresponds to a specific numerical value of the slope, namely two minutes of finishing time per minute of halfway split. The usual regression null hypothesis of a zero slope is uninformative in this setting, because finishing time must increase with the halfway split. Testing the fitted slope against two, rather than against zero, turns a routine regression into a direct test of whether a mass-participation field paces evenly.

This paper fits that model to the full finishing field of the 2023 Boston Marathon. The estimated slope is 2.218 rather than 2, a difference that is far larger than its standard error, so each extra minute taken to reach halfway is associated with about 1.22 extra minutes over the second half instead of the one minute even pacing would require. The halfway split explains almost all of the variation in finishing time, yet the remaining spread is wide enough in practical terms that a prediction for an individual runner carries an uncertainty of roughly 19 minutes in either direction. The rest of the paper is organized as follows. Section 2 describes the data set, its variables, and its limitations. Section 3 sets out the regression model, the assumptions behind it, and the checks used to validate it. Section 4 reports the fitted model and its practical implications, and Section 5 summarizes the findings, the weaknesses of the study, and ways it could be extended.

2 Data

2.1 Source and observational units

The data are the official results of the 2023 Boston Marathon, run on April 17, 2023. The file was compiled by Fay, Dykstra, and Ramler (2023) from results published by the Boston Athletic Association (Boston Athletic Association 2023) and is distributed through the SCORE Network Sports Data Repository. Each row is one runner who finished the race, so the observational unit is a single finisher, and the data set covers the complete finishing field of 26,598 runners rather than a sample from it. Runners who started but did not finish never appear in the results, a point I return to below.

2.2 Variables

The file contains 15 columns recording each finisher’s identity, age group, sex, finishing places, and times. Four are used here.

The response variable is **net finishing time**, the elapsed time from the moment a runner crosses the starting mat to the moment they cross the finishing mat, recorded by a timing chip and converted here from seconds to minutes. Boston starts runners in waves and corrals, so the starting gun and a given runner’s actual start can differ considerably: across all finishers the gun time exceeds the net time by a median of 2.5 minutes and by as much as 27.1 minutes. Net time measures the runner’s own race and is the appropriate response for a question about pacing; gun time would mix in corral assignment, which reflects qualifying time rather than what happened on the day.

The explanatory variable is the **half-marathon split**, the net elapsed time recorded at the halfway timing mat, also converted to minutes. From these two I derive **second-half time**, the difference between the finishing time and the split, and the **split ratio**, the second-half time divided by the split. A ratio above one means the runner covered the second half more slowly than the first. **Sex**

(recorded in the source file as **M** or **W**) and **age group** (ten bands from 18–39 to 80+) are used for description only; they are not part of the fitted model.

Both times come from chip timing, which is accurate to about one second. That error is negligible next to a standard deviation of 19.3 minutes in the split, so measurement error in either variable can be ignored. One structural feature of the variables does matter and is addressed in Section 3: the finishing time contains the halfway split, because a marathon is the sum of its two halves.

2.3 Missing data and field characteristics

72 finishers (0.27% of the field) have no recorded half-marathon split, most likely because of a missed chip read at the mat. Because the split is the explanatory variable, these runners cannot enter the model and are dropped, leaving 26,526 runners. Their exclusion is unlikely to distort the results: their mean finishing time, 224.4 minutes, is close to the 222.5 minutes of the runners retained, and they are spread across sexes and age groups in much the same proportions as the rest of the field. No other variable used here has missing values, and no runner appears twice.

Table 1 summarizes the analysis variables. The field is wide ranging: halfway splits run from 1:02:19 to 3:27:15 and finishing times from 2:05:54 to 7:20:31. The median split ratio of 1.058 says that a typical finisher ran the second half about 5.8% slower than the first.

Table 1: Median times (in minutes) and pacing measures for the 26,526 finishers of the 2023 Boston Marathon with a recorded half-marathon split, for the whole field and separately by sex. The split ratio is second-half time divided by half-marathon split; a value above 1 indicates a slower second half. The final column gives the percentage of runners in each group whose second half was slower than their first. Women were slower overall than men in this field but slowed by a similar proportion.

Group	n	Half split	Finish	Second half	Split ratio	Slower 2nd half (%)
All finishers	26,526	103.5	213.9	110.1	1.058	89.0
Women	11,384	109.5	226.1	116.1	1.058	88.7
Men	15,142	97.1	200.9	103.7	1.058	89.3

Two features of the field shape how far the results travel. First, Boston is not an open race: most entrants qualify by running a time standard at an earlier marathon, and the remainder enter through charity programs. The field is therefore faster and more experienced than a typical city marathon, and if experience protects against fading, the slowdown measured here is likely to be milder than it would be in an open field. Second, every runner in these data raced the same course on the same day, and ambient temperature is known to affect how much runners slow, with slower runners most affected (Ely et al. 2008). Because conditions do not vary within the data set, their effect cannot be separated from pacing behaviour; whatever the weather on the day contributed is absorbed into the estimated relationship.

Figure 1 shows the distributions of the two modelled variables and of the split ratio. Both time distributions are right skewed, with a long tail of slower finishers. The split ratio is centred a little above one and has a pronounced right tail: 89.0% of finishers ran a slower second half, and a small group took more than twice as long over the second half as the first.

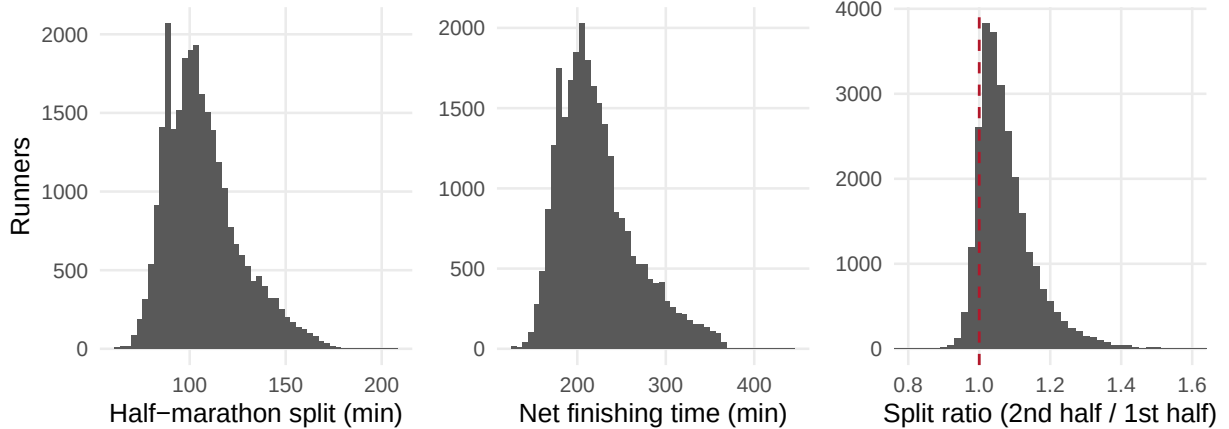


Figure 1: Distributions of the two modelled variables and of the split ratio for 26,526 finishers. Half-marathon split (left) and net finishing time (centre) are both right skewed. The split ratio (right, second-half time divided by half-marathon time) is centred just above the dashed even-pacing line at 1.0, showing that a slower second half was the norm: 89.0% of finishers ran a positive split. The horizontal axis of the right panel is restricted to 0.8–1.6 for legibility, which hides 0.15% of runners whose pacing was more extreme still; the most uneven runner took more than twice as long over the second half as over the first.

2.4 Additional and alternative data

The published results contain richer pacing information than this file includes: the Boston Athletic Association records splits at 5 km intervals as well as at halfway, which would allow the shape of the slowdown to be traced rather than summarized by a single second-half figure. Results for other editions of the race are published in the same format, and other large marathons publish comparable files, so the analysis could be repeated across years and courses. Hourly weather observations for the course are available from the National Oceanic and Atmospheric Administration and could be matched to each runner’s time on the course, which would help separate heat effects from pacing behaviour. Qualifying times, which the results do not include, would give a measure of each runner’s recent fitness and so distinguish a runner who paced badly from one who was simply having a poor day. None of these sources is needed for the question asked here, but each would support the extensions discussed in Section 5.

3 Methods

3.1 The model

Let Y_i denote the net finishing time in minutes of runner i and let x_i denote that runner’s half-marathon split in minutes, for $i = 1, \dots, n$ with $n = 26,526$. The simple linear regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2).$$

Here β_0 is the intercept, the expected finishing time for a runner with a halfway split of zero minutes; β_1 is the slope, the expected increase in finishing time associated with a one-minute increase in

the halfway split; ε_i is the error for runner i , the amount by which that runner's actual finishing time departs from the mean for runners with the same split; and σ^2 is the variance of those errors, assumed the same for every runner. The notation $\stackrel{iid}{\sim}$ means the errors are independent of one another and drawn from the same normal distribution.

The parameters are estimated by ordinary least squares, which chooses the values $\hat{\beta}_0$ and $\hat{\beta}_1$ minimizing the sum of squared vertical distances between the observed finishing times and the fitted line, $\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$. The residual standard deviation $\hat{\sigma}$ estimates the typical size of those distances and is reported in minutes, which makes it directly readable as the precision of a prediction.

A single explanatory variable is appropriate because the question is itself a one-variable question: it asks what a runner can infer at the halfway mat, when the split is the only new information available. Sex, age, and weather all influence finishing time, but adding them would answer a different question and would produce an equation a runner could not evaluate from the clock.

3.2 The even-pacing benchmark

If every runner covered the two halves of the course in equal times, then $Y_i = 2x_i$ exactly, which corresponds to $\beta_0 = 0$ and $\beta_1 = 2$. The slope therefore has a meaningful benchmark value, and the hypotheses of interest are

$$H_0 : \beta_1 = 2 \quad \text{versus} \quad H_1 : \beta_1 \neq 2,$$

tested with the statistic $t = (\hat{\beta}_1 - 2)/SE(\hat{\beta}_1)$ compared to a t distribution with $n - 2$ degrees of freedom. A slope above two means that a one-minute increase in the halfway split is associated with more than one extra minute in the second half, so that runners who reach halfway later also give up proportionally more time afterwards. The conventional test of $H_0 : \beta_1 = 0$ is reported by software automatically but carries no information here, since finishing time cannot fail to increase with the halfway split.

The same question can be posed by regressing second-half time on the halfway split. Because second-half time is $Y_i - x_i$, that model has slope $\beta_1 - 1$ and identical standard errors, so it tests the even-pacing benchmark of one with exactly the same t statistic. The choice between the two framings is presentational, and I use finishing time as the response because it is the quantity a runner wants to predict.

One consequence of this structure deserves emphasis. The response contains the explanatory variable by construction, since a finishing time is the halfway split plus the second-half time. A large R^2 is therefore guaranteed and should not be read as evidence that pacing is well understood; it mostly reflects the arithmetic. The informative quantities are the distance of the slope from two, which is not mechanical, and the residual standard deviation, which measures how precisely the second half can be anticipated.

3.3 Assumptions and diagnostics

Four assumptions support the model, and I checked each.

Linearity. The mean finishing time is assumed to change at a constant rate across the range of halfway splits. A scatterplot with the fitted line and a plot of residuals against fitted values were

used to look for curvature.

Independence. Errors are assumed unrelated across runners. This is the weakest assumption in the analysis: runners race in packs, follow pacing groups, and share the same weather and course conditions, so two runners near each other are likely to fade together. Clustering of this kind leaves the slope estimate unbiased but can make standard errors too small. With a confidence interval already narrow enough that the conclusion does not turn on its width, the practical risk is limited, but the assumption cannot be verified from the results file alone.

Constant variance. Residual spread should not change with the fitted value. It plainly does here, so this assumption fails; the response is described below.

Normality. The errors are assumed normally distributed. A normal quantile-quantile plot shows heavy tails, and the residuals are right skewed. For inference about β_1 this matters little, because with tens of thousands of observations the sampling distribution of the slope is approximately normal regardless. It matters more for prediction intervals for an individual runner, which rely on the shape of the error distribution rather than on large-sample behaviour, so those intervals are reported as approximations.

Influence was assessed with Cook’s distance, which measures how much the fitted line would move if a single runner were removed. The largest value in the data is 0.060, far below the conventional threshold of one, so no individual result drives the fit. Unusual pacing patterns, including runners who walked large parts of the first half and those who collapsed in the second, were deliberately retained: they are genuine race outcomes rather than recording errors, and removing them would bias the estimated slowdown toward zero.

3.4 Handling the failure of constant variance

Because residual spread grows with the halfway split, the least squares estimates remain unbiased but the usual standard errors are not trustworthy. I addressed this in two ways, which together serve as the validation step for the model.

First, I recomputed the standard error of the slope using a heteroskedasticity-consistent (HC1) estimator (Zeileis 2004), which does not assume constant variance. Second, I refitted the model with both variables on the natural log scale,

$$\log(Y_i) = \gamma_0 + \gamma_1 \log(x_i) + \eta_i,$$

where γ_1 is the elasticity of finishing time with respect to the halfway split: the approximate percentage change in finishing time associated with a 1% increase in the split. Even pacing implies $\log(Y_i) = \log 2 + \log(x_i)$, so the benchmark for γ_1 is one rather than two, and a value above one again means that slower runners fade proportionally more. Taking logarithms compresses the long right tail of both variables and is the standard remedy when spread grows with level. If the conclusion about pacing survives both changes, it does not depend on the assumption that failed. Formal evidence for non-constant variance is provided by the Breusch–Pagan test (Breusch and Pagan 1979; Zeileis and Hothorn 2002).

3.5 Possible extensions to the model

Three extensions would change the analysis in predictable ways. Adding sex and age group as further explanatory variables, with interactions against the halfway split, would turn the model into

a multiple regression and would let the slope differ across groups; since men and women in this field slowed by almost the same proportion (Table 1), this is likely to sharpen the intercept more than the slope, and it would cost the equation its usability at the mat. Allowing curvature, through a quadratic term or a spline in the split, would relax the constant-rate assumption; the residual plot shows little curvature, so the fitted values would move only slightly, but any extrapolation beyond the observed range would become less stable. Weighted least squares or quantile regression would model the growing spread directly rather than correcting for it after the fact; the slope estimate would change little, because the robust and log-log refits already leave it intact, but the prediction intervals would widen for slower runners and narrow for faster ones, which is where the present intervals are least accurate. None of these extensions is required to answer the question posed here, and each is set out more fully in Section 5.

3.6 Software

All work was carried out in R version 4.3.3 (R Core Team 2024). Data handling used **dplyr** (Wickham et al. 2023) and **readr** (Wickham, Hester, and Bryan 2024); figures were produced with **ggplot2** (Wickham 2016) and assembled with **patchwork** (Pedersen 2024); robust standard errors and the Breusch–Pagan test came from **sandwich** (Zeileis, Köll, and Graham 2020; Zeileis 2004) and **lmtest** (Zeileis and Hothorn 2002); and the report was generated with **knitr** (Xie 2023), **rmarkdown** (Xie, Allaire, and Golemund 2018), **bookdown** (Xie 2016), and **kableExtra** (Zhu 2024). Models were fitted with the `lm` function in R’s built-in **stats** package.

4 Results

4.1 The fitted model

The fitted regression line is

$$\widehat{\text{finish}} = -14.53 + 2.218 \times \text{half-marathon split},$$

with both times in minutes. Table 2 reports the estimates.

Table 2: Ordinary least squares estimates for the regression of net finishing time (minutes) on half-marathon split (minutes), fitted to 26,526 finishers of the 2023 Boston Marathon. Confidence intervals are 95% intervals based on constant-variance standard errors. The slope of 2.218 exceeds the value of 2 implied by even pacing, and the interval for the slope excludes 2 by a wide margin.

Term	Estimate	Std. error	CI low	CI high
Intercept	-14.526	0.327	-15.167	-13.886
Half-marathon split	2.218	0.00301	2.212	2.224

The slope of 2.218 is the central result. A runner who reaches halfway one minute later than another runner is expected to finish 2.22 minutes later, not the two minutes that equal halves would imply. Rearranged, each extra minute taken to reach halfway is associated with 1.22 extra minutes over the second half. Comparing the estimate with the even-pacing benchmark gives $t = 72.4$ on 26,524 degrees of freedom, with a p -value below 0.001, so a field that paces evenly on average is not consistent with these results. The 95% confidence interval for the slope, 2.212 to 2.224, lies entirely above two.

The intercept of -14.53 minutes has no physical meaning on its own: a halfway split of zero minutes is impossible, and the observed splits range from 62.3 to 207.2 minutes, so a split of zero is far outside the data. Its role is to position the line over the range where runners actually are. Its negative value is what allows a line with a slope steeper than two to pass through the middle of a field in which typical finishing times are close to twice the halfway split.

The model accounts for 95.3% of the variation in finishing times, and the correlation between the two times is 0.976. As explained in Section 3, these figures are inflated by the fact that finishing time contains the halfway split, and they should not be read as a measure of how well pacing is understood. The residual standard deviation of 9.43 minutes is the more useful summary of fit: the typical runner finishes within about 9 minutes of the prediction made from their halfway split.

Figure 2 shows the data with the fitted line and the even-pacing line $y = 2x$ for comparison. The fitted line sits above the even-pacing line across the whole range and pulls further above it as splits lengthen, which is the visual signature of a slope greater than two.

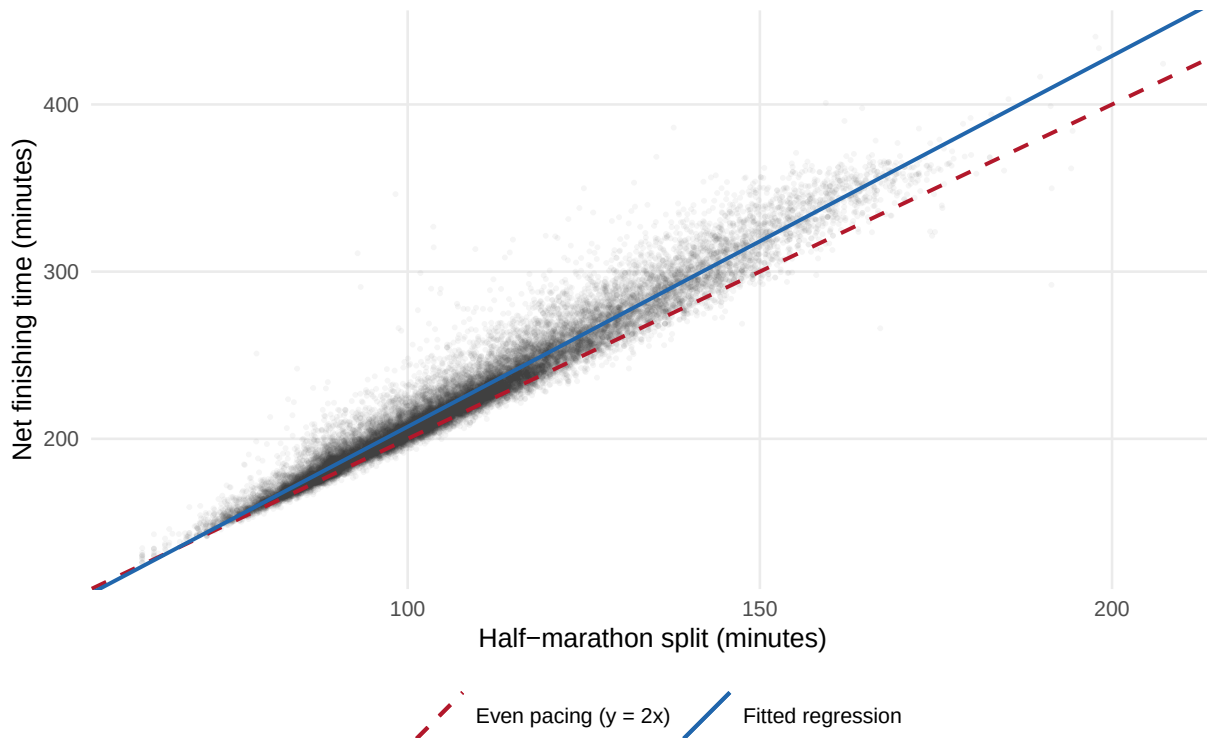


Figure 2: Net finishing time against half-marathon split for 26,526 finishers of the 2023 Boston Marathon, with each point one runner (plotted with transparency so that darker regions indicate more runners). The solid line is the fitted regression, $\text{finish} = -14.53 + 2.218 \times \text{split}$; the dashed line is the even-pacing reference $\text{finish} = 2 \times \text{split}$. The fitted line lies above the reference line throughout and diverges from it as splits lengthen, showing that runners who reach halfway later lose proportionally more time over the second half. Points above the dashed line are runners who slowed in the second half.

4.2 What the model means for a runner

Table 3 converts the fitted line into the projections a runner might make at the mat. For each halfway split it gives the finishing time the doubling rule predicts, the finishing time the model predicts, and an approximate 95% prediction interval for an individual runner.

Table 3: Finishing times projected from five halfway splits, comparing the doubling rule with the fitted regression. The final column gives the number of minutes by which doubling understates the model prediction; it grows with the halfway split, from about 5 minutes for a 1:30 split to about 18 minutes for a 2:30 split. Prediction intervals are approximate because the residuals are heavy tailed, and they are wide – roughly 37 minutes across – which shows that the halfway split pins down the average finishing time far more precisely than any individual runner’s.

Halfway split	Doubling rule	Model prediction	95% prediction interval	Shortfall of doubling
1:30:00	3:00:00	3:05:05	2:46:35 to 3:23:34	5.1 min
1:45:00	3:30:00	3:38:21	3:19:51 to 3:56:50	8.3 min
2:00:00	4:00:00	4:11:37	3:53:07 to 4:30:06	11.6 min
2:15:00	4:30:00	4:44:53	4:26:23 to 5:03:22	14.9 min
2:30:00	5:00:00	5:18:09	4:59:39 to 5:36:38	18.1 min

The shortfall of the doubling rule grows across the table, from about 5 minutes for a runner passing halfway in 1:30 to about 18 minutes for a runner passing in 2:30. This is the practical form of the result: the faster the runner, the better the doubling rule works. For the median finisher, who reached halfway in 1:43:33, doubling understates the finish by about 8 minutes, and the shortfall reaches a quarter of an hour only for the slowest 10% of the field. The prediction intervals, however, are wide: about 37 minutes from bottom to top. The regression describes the average behaviour of the field accurately while leaving a great deal of individual variation unexplained, which is what one would expect given that the second half of a marathon depends on training, fuelling, heat tolerance, and luck that the halfway clock cannot observe.

The descriptive pacing figures agree with the model. 89.0% of finishers ran a slower second half than first, and the median runner’s second half was 5.8% slower than their first.

4.3 Model checking

Figure 3 presents the diagnostics described in Section 3. The residual plot is roughly flat, so the straight-line form is adequate, but the band of residuals widens markedly from left to right. The residual standard deviation rises from 5.9 minutes among the fastest tenth of the field to 14.1 minutes among the slowest, a 2.4-fold increase, and the Breusch–Pagan test rejects constant variance decisively ($p < 0.001$). The quantile-quantile plot shows approximate normality through the centre with heavy tails at both ends, the right tail reflecting runners who faded badly.

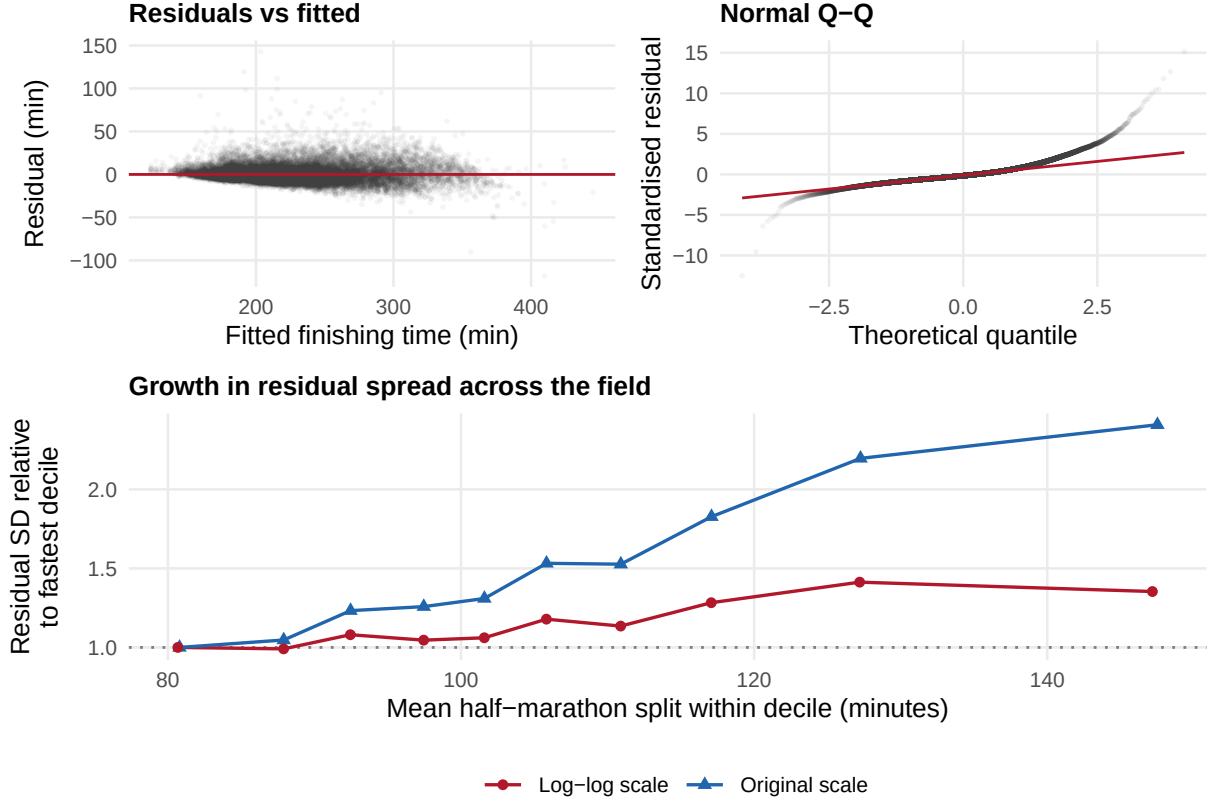


Figure 3: Regression diagnostics for the fitted model. Top left: residuals against fitted values, flat on average (so a straight line is adequate) but fanning out as fitted times increase. Top right: normal quantile-quantile plot of standardised residuals, close to the reference line through the centre but with heavy tails at both ends. Bottom: residual standard deviation within tenths of the half-marathon split, expressed relative to the fastest decile so that the two fits can be compared on one scale. On the original scale the spread grows 2.4-fold across the field, which is why the constant-variance assumption fails; on the log-log scale it grows only 1.4-fold, which is why that fit is used as a sensitivity check.

Table 4 shows that the conclusion is unaffected by either remedy. Robust standard errors roughly 1.3 times the constant-variance ones still place the slope far above two ($t = 56.3$). On the log-log scale the elasticity is 1.058, again above the even-pacing benchmark of one ($t = 42.5$), and the residual spread across deciles rises by a factor of only 1.4 rather than 2.4, so the transformation does most of the work of stabilising the variance. Both versions of the model say the same thing: runners who reach halfway later give up proportionally more time in the second half.

5 Discussion

Across the full finishing field of the 2023 Boston Marathon, finishing time rises by 2.22 minutes for each additional minute of half-marathon split, rather than the two minutes that even pacing implies. The doubling rule used at the halfway mat is therefore biased, and predictably so: it is nearly right for the fastest runners, understates the finish by about 8 minutes for the median finisher, and

Table 4: Sensitivity of the pacing conclusion to the failure of the constant-variance assumption. Each row tests the fitted slope against the value implied by even pacing (2 on the original scale, 1 on the log-log scale). Robust standard errors widen the interval slightly and the log-log model changes the scale entirely, but in all three versions the slope sits far above its even-pacing benchmark, so the conclusion does not depend on the assumption that failed.

Model	Slope	Std. error	Benchmark	t statistic
Original scale, constant-variance SE	2.218	0.0030	2	72.4
Original scale, robust (HC1) SE	2.218	0.0039	2	56.3
Log-log scale, constant-variance SE	1.058	0.0014	1	42.5

is optimistic by a quarter of an hour or more for the slowest 10% of the field. Replacing it with the fitted line, or with the rough mental version of doubling the split and adding a further fifth of it, removes that bias. The wide prediction intervals are as important as the slope: the halfway split fixes the field’s average behaviour tightly but leaves an individual runner’s finish uncertain by roughly 9 minutes on average and by about 19 minutes either side at the 95% level, so the projection should be treated as a central estimate rather than a forecast. That slower runners fade more is consistent with previous findings from other marathons (March et al. 2011; Santos-Lozano et al. 2014), and this analysis adds a single directly usable equation for a complete mass-participation field.

Several weaknesses qualify these conclusions. The data describe finishers only. Runners who started and dropped out are absent from the results, and they are disproportionately the runners whose second half went worst, so the true average slowdown in the starting field is steeper than the estimate here: the result is a lower bound on fading. The analysis covers one race on one day, and Boston’s qualifying standards make its field faster and more experienced than an open marathon, so the numerical slope should not be transferred to a different race without refitting. Weather on the day is baked into the estimate and cannot be separated from it. The constant-variance assumption fails outright and the residuals are heavy tailed; robust standard errors and a log-log refit leave the conclusion intact, but the prediction intervals remain approximate and are probably too narrow for the slowest runners and too wide for the fastest. Runners influence each other’s pacing, which the independence assumption ignores. Finally, the model is descriptive, not causal: it does not show that starting faster causes a better finish, since runners who reach halfway quickly are fitter, and fitness rather than pace choice explains much of the association.

The natural extensions follow from those limitations. Adding sex, age group, and their interactions with the halfway split would test whether the slowdown differs across runners in the way earlier work reports (Deaner et al. 2015), at the cost of an equation a runner could no longer evaluate from the clock alone. Fitting the same model to several editions of the race, with a weather variable for each day, would separate heat effects from pacing behaviour and would show whether the slope is a stable property of the field (Ely et al. 2008). The 5 km splits published by the organizers would allow the slowdown to be traced through the race rather than summarized once, which is the natural way to study late-race collapse (Smyth 2021). Quantile regression would model the spread directly instead of treating it as a nuisance, and would give runners prediction limits that widen appropriately with their split. Each of these would refine the estimate; none is likely to overturn the central finding, which is that a marathon field does not pace evenly, and that the shortfall of the doubling rule grows with the time on the clock at halfway.

References

- Boston Athletic Association. 2023. “2023 Boston Marathon Official Results.” Boston Athletic Association. <https://www.baa.org>.
- Breusch, T. S., and A. R. Pagan. 1979. “A Simple Test for Heteroscedasticity and Random Coefficient Variation.” *Econometrica* 47 (5): 1287–94. <https://doi.org/10.2307/1911963>.
- Deaner, Robert O., Rickey E. Carter, Michael J. Joyner, and Sandra K. Hunter. 2015. “Men Are More Likely Than Women to Slow in the Marathon.” *Medicine & Science in Sports & Exercise* 47 (3): 607–16. <https://doi.org/10.1249/MSS.0000000000000432>.
- Ely, Matthew R., David E. Martin, Samuel N. Cheuvront, and Scott J. Montain. 2008. “Effect of Ambient Temperature on Marathon Pacing Is Dependent on Runner Ability.” *Medicine & Science in Sports & Exercise* 40 (9): 1675–80. <https://doi.org/10.1249/MSS.0b013e3181788da9>.
- Fay, Jack, A. J. Dykstra, and Ivan Ramler. 2023. “2023 Boston Marathon Runners.” SCORE Network Sports Data Repository. https://data.scorenetwork.org/running/boston_marathon_2023.html.
- March, Daniel S., Paul M. Vanderburgh, Peter J. Titlebaum, and Meghan L. Hoops. 2011. “Age, Sex, and Finish Time as Determinants of Pacing in the Marathon.” *Journal of Strength and Conditioning Research* 25 (2): 386–91. <https://doi.org/10.1519/JSC.0b013e3181bffd0f>.
- Pedersen, Thomas Lin. 2024. *patchwork: The Composer of Plots*. <https://patchwork.data-imaginist.com>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Santos-Lozano, Alejandro, Pilar S. Collado, Carl Foster, Alejandro Lucia, and Nuria Garatachea. 2014. “Influence of Sex and Level on Marathon Pacing Strategy. Insights from the New York City Race.” *International Journal of Sports Medicine* 35 (11): 933–38. <https://doi.org/10.1055/s-0034-1367048>.
- Smyth, Barry. 2021. “How Recreational Marathon Runners Hit the Wall: A Large-Scale Data Analysis of Late-Race Pacing Collapse in the Marathon.” *PLoS ONE* 16 (5): e0251513. <https://doi.org/10.1371/journal.pone.0251513>.
- Wickham, Hadley. 2016. *ggplot2: Elegant Graphics for Data Analysis*. 2nd ed. Springer-Verlag New York. <https://ggplot2.tidyverse.org>.
- Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2023. *dplyr: A Grammar of Data Manipulation*. <https://dplyr.tidyverse.org>.
- Wickham, Hadley, Jim Hester, and Jennifer Bryan. 2024. *readr: Read Rectangular Text Data*. <https://readr.tidyverse.org>.
- Xie, Yihui. 2016. *bookdown: Authoring Books and Technical Documents with R Markdown*. Boca Raton, Florida: Chapman and Hall/CRC. <https://bookdown.org/yihui/bookdown>.
- . 2023. *knitr: A General-Purpose Package for Dynamic Report Generation in R*. <https://yihui.org/knitr/>.
- Xie, Yihui, J. J. Allaire, and Garrett Golemund. 2018. *R Markdown: The Definitive Guide*. Boca Raton, Florida: Chapman and Hall/CRC. <https://bookdown.org/yihui/rmarkdown>.
- Zeileis, Achim. 2004. “Econometric Computing with HC and HAC Covariance Matrix Estimators.” *Journal of Statistical Software* 11 (10): 1–17. <https://doi.org/10.18637/jss.v011.i10>.
- Zeileis, Achim, and Torsten Hothorn. 2002. “Diagnostic Checking in Regression Relationships.” *R News* 2 (3): 7–10. <https://CRAN.R-project.org/doc/Rnews/>.
- Zeileis, Achim, Susanne Köll, and Nathaniel Graham. 2020. “Various Versatile Variances: An Object-Oriented Implementation of Clustered Covariances in R.” *Journal of Statistical Software* 95 (1): 1–36. <https://doi.org/10.18637/jss.v095.i01>.

Zhu, Hao. 2024. *kableExtra: Construct Complex Table with 'kable' and Pipe Syntax*. <https://haozhu233.github.io/kableExtra/>.